

Average rate of change



1

a 37°C

b 1.3°C/min

c $\frac{dT}{dt} = 1.4 - 0.04t$

d 1.2°C/min

e $t = 35$

2

a \$54

b \$8.33 per year

c $\frac{dP}{dx} = x^2 - 16x + 63$

d \$15 per year

e $x = 7, x = 9$

3

a $A'(t) = \frac{8}{5} + \frac{4t}{25}$

b cm^2/h

c $2.8\text{ cm}^2/\text{h}$

d $A'(t) = 2.56$

4

a $-6000v^{-2}$

b Beats per minute per mL

c 75

d -0.75 bpm/mL

e -0.9375 bpm/mL

Rates of change

5 $\frac{dR}{dt} = \frac{1}{17} + \frac{t}{54}$

6 $\frac{dV}{dP} = -kP^{-2}$

7

a $V' = -24x^5 - 25x^4 + 1000x^3$

b $-120\,393$

8

a $\frac{dR}{dp} = -6p + 45$

b 21

c -3

d \$4, because they will make a revenue of \$21, instead of losing \$3.

9

$$a \quad \frac{dy}{dx} = 25x^{-\frac{1}{2}} + \frac{40}{3}x^{-\frac{2}{3}} - \frac{1}{4}x$$

c Increase



$$b \quad \frac{dy}{dx} = 7 \text{ kWh/h}$$

10

$$a \quad L' = \frac{1}{1000} (4t^3 - 24t^2 + 850t - 1300)$$

$$b \quad L' = \frac{41\,443}{500}$$

c Going in

11

$$a \quad \frac{3n^2}{2} - 410$$

b 17 740 m² per animalc 204 940 m² per animal

d No, because as the population increases, the area rate increases.

12

$$a \quad \frac{dV}{dt} = 1000 (-3t^2 + 150t - 1875)$$

$$b \quad 75\,000 \text{ L/s}$$

$$c \quad t = 15, t = 35$$

13

$$a \quad f'(x) = -16\,000x^{-3} - 4000x^{-2}$$

b How much the voltage generated changes per million kilometres when the probe is a certain distance from the sun.

$$c \quad -0.0064 \text{ mV/Gm}$$

14

$$a \quad P' = 32x^{-\frac{1}{2}} + \frac{80}{3}x^{-\frac{2}{3}}$$

b i 15 mice/day ii 7 mice/day iii 5 mice/day

c The growth is increasing but the rate of increase is slowing down.

15

$$a \quad \frac{dr}{dt} = 1.5t^{-\frac{2}{3}}$$

b r increases more slowly.

$$c \quad V = 121.5\pi t$$

$$d \quad \frac{dV}{dt} = 121.5\pi$$

$$e \quad 121.5\pi \text{ cm}^3/\text{s}$$

f The volume increases at a constant rate.

16

$$\text{a} \quad \frac{dE}{dg} = 400g - \frac{3g^2}{2}$$

$$\text{c} \quad 133\frac{1}{3} \text{ units}$$



$$\text{b} \quad 10\,650$$

17

$$\text{a} \quad V = \frac{9\pi t^3}{125}$$

$$\text{c} \quad \frac{432\pi}{125} \text{ units}^3/\text{s}$$

$$\text{b} \quad \frac{dV}{dt} = \frac{27\pi t^2}{125}$$

18

$$\text{a} \quad \$5423$$

$$\text{c} \quad \frac{dC}{dx} = 7 + 60x^{-\frac{1}{2}}$$

$$\text{b} \quad \$11 \text{ per item}$$

$$\text{d} \quad \$9 \text{ per item}$$

19

$$\text{a} \quad t + \frac{9}{10} \text{ m}^2/\text{day}$$

$$\text{b} \quad \frac{209}{10} \text{ m}^2/\text{day}$$

$$\text{c} \quad t = 38$$

20

$$\text{a} \quad M' = 3x^2 + 36x + 81$$

$$\text{c} \quad 221.43 \text{ g}$$

$$\text{b} \quad \text{g/cm}$$